Testing of Hypothesis and Confidence Interval ?

When to Use What?

Technique	Use When
T-Test	Comparing means of 2 groups
ANOVA	Comparing means of 3+ groups
Chi-Square Test	Comparing categorical variables
Confidence Interval	Estimating a population parameter with uncertainty

Problem Statements with Hypothesis Testing

Q1. Do Male & Female customers spend the same amount?

→ Two-Sample T-Test

```
from scipy import stats

male = df[df["Gender"] == "Male"]["Purchase Amount (USD)"]
female = df[df["Gender"] == "Female"]["Purchase Amount (USD)"]

t_stat, p_value = stats.ttest_ind(male, female)

print(f"T-statistic: {t_stat:.4f}")

print(f"P-value: {p_value:.4f}")

if p_value < 0.05:

print(" Reject H0 → Significant difference in spending between genders")

else:
```

print(" Fail to Reject H0 → No significant difference")

Hypothesis :-

H0 (Null)   : Mean spend of Male = Mean spend of Female

H1 (Alternate) : Mean spend of Male ≠ Mean spend of Female

Q2. Does purchase amount differ across all 4 seasons?

→ One-Way ANOVA

```
spring = df[df["Season"] == "Spring"]["Purchase Amount (USD)"]

summer = df[df["Season"] == "Summer"]["Purchase Amount (USD)"]

fall   = df[df["Season"] == "Fall"]["Purchase Amount (USD)"]

winter = df[df["Season"] == "Winter"]["Purchase Amount (USD)"]

f_stat, p_value = stats.f_oneway(spring, summer, fall, winter)

print(f"F-statistic: {f_stat:.4f}")

print(f"P-value: {p_value:.4f}")

if p_value < 0.05:

   print(" Reject H0 → Season significantly affects purchase amount")

else:

   print(" Fail to Reject H0 → No significant difference across seasons")
```

H0 : Mean purchase amount is equal across all seasons

H1 : At least one season has a different mean

Q3. Do subscribers spend more than non-subscribers?

→ One-Tailed T-Test

```
subscribed     = df[df["Subscription Status"] == "Yes"]["Purchase Amount (USD)"]

not_subscribed = df[df["Subscription Status"] == "No"]["Purchase Amount (USD)"]

t_stat, p_value = stats.ttest_ind(subscribed, not_subscribed, alternative="greater")

print(f"T-statistic: {t_stat:.4f}")

print(f"P-value: {p_value:.4f}")

if p_value < 0.05:

   print("Reject H0 → Subscribers spend significantly MORE")

else:

   print("Fail to Reject H0 → No significant difference")
```

H0 : Subscribers spend ≤ Non-subscribers

H1 : Subscribers spend > Non-subscribers

Q4. Is Gender independent of Payment Method preference?

→ Chi-Square Test

```
contingency_table = pd.crosstab(df["Gender"], df["Payment Method"])

print(contingency_table)

chi2, p_value, dof, expected = stats.chi2_contingency(contingency_table)

print(f"\nChi2 Statistic: {chi2:.4f}")

print(f"P-value: {p_value:.4f}")

print(f"Degrees of Freedom: {dof}")

if p_value < 0.05:

    print(" Reject H0 → Gender & Payment Method ARE related")

else:

    print(" Fail to Reject H0 → Gender & Payment Method are INDEPENDENT")
```

H0 : Gender and Payment Method are independent

H1 : Gender and Payment Method are related

Q5. Is Discount usage related to Frequency of Purchase?

→ Chi-Square Test

```
ct = pd.crosstab(df["Discount Applied"], df["Frequency of Purchases"])

chi2, p_value, dof, expected = stats.chi2_contingency(ct)

print(f"Chi2: {chi2:.4f}, P-value: {p_value:.4f}")

if p_value < 0.05:

  print(" Discount usage IS related to purchase frequency")

else:

  print(" No relationship found")
```

Confidence Interval :-

CI-1. 95% CI for Average Purchase Amount

```
import numpy as np

data = df["Purchase Amount (USD)"].dropna()

n = len(data)

mean = np.mean(data)

se = stats.sem(data)  # Standard Error

ci = stats.t.interval(0.95, df=n-1, loc=mean, scale=se)

print(f"Mean Purchase Amount : ${mean:.2f}")

print(f"95% Confidence Interval : ${ci[0]:.2f} to ${ci[1]:.2f}")
```

Meaning: We are **95% confident** the true average purchase amount of ALL customers falls in this range.

CI-2. 95% CI — Male vs Female Spending

```
for gender in ["Male", "Female"]:
    data = df[df["Gender"] == gender]["Purchase Amount (USD)"].dropna()
    mean = np.mean(data)
    ci = stats.t.interval(0.95, df=len(data)-1, loc=mean, scale=stats.sem(data))
    print(f"{gender} → Mean: ${mean:.2f} | 95% CI: (${ci[0]:.2f}, ${ci[1]:.2f})")
```

CI-3. 95% CI for Average Review Rating

```
data = df["Review Rating"].dropna()
mean = np.mean(data)
ci = stats.t.interval(0.95, df=len(data)-1, loc=mean, scale=stats.sem(data))
print(f"Mean Rating : {mean:.2f}")
print(f"95% CI : ({ci[0]:.2f}, {ci[1]:.2f})")
```